

1 Gaussian Quadrature

The Newton-Cotes formulas in Section 4 of Module 4 were derived by integrating interpolating polynomials. The error term in the interpolating polynomial of degree n involves the $(n + 1)$ st derivative of the function being approximated, so a Newton-Cotes formula is exact when approximating the integral of any polynomial of degree less than or equal to n .

All the Newton-Cotes formulas use values of the function at equally-spaced points. This restriction is convenient when the formulas are combined to form the composite rules we considered in Section 1 of this Module, but it can significantly decrease the accuracy of the approximation. Consider, for example, the Trapezoidal rule applied to determine the integrals of the functions whose graphs are shown in Figure 1.

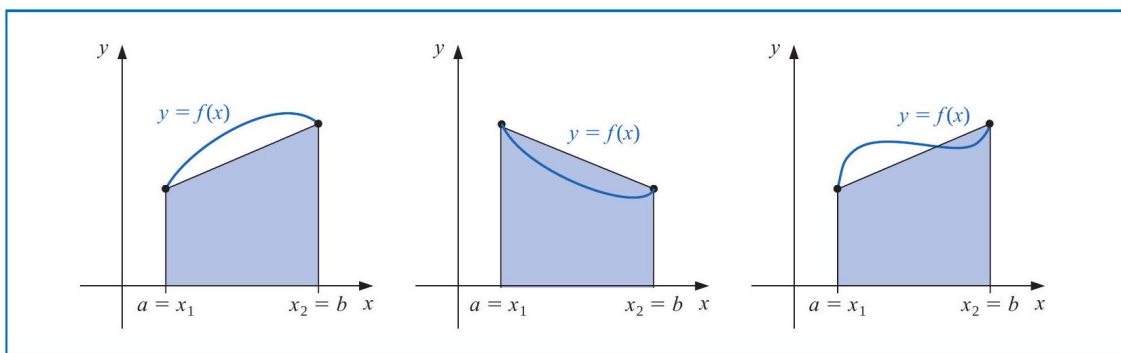


Figure 1: Figure 4

The Trapezoidal rule approximates the integral of the function by integrating the linear function that joins the endpoints of the graph of the function. But this is not likely the best line for approximating the integral. Lines such as those shown in Figure 2 would likely give much better approximations in most cases.

Gaussian quadrature chooses the points for evaluation in an optimal, rather than equally spaced, way. The nodes $x_1, x_2, \dots, x_n$ in the interval $[a, b]$ and coefficients $c_1, c_2, \dots, c_n$, are chosen to minimize the expected error obtained in the approximation

$$\int_a^b f(x)dx \approx \sum_{i=1}^n c_i f(x_i)$$

To measure this accuracy, we assume that the best choice of these values produces the exact result for the largest class of polynomials, that is, the choice that gives the greatest degree of precision.

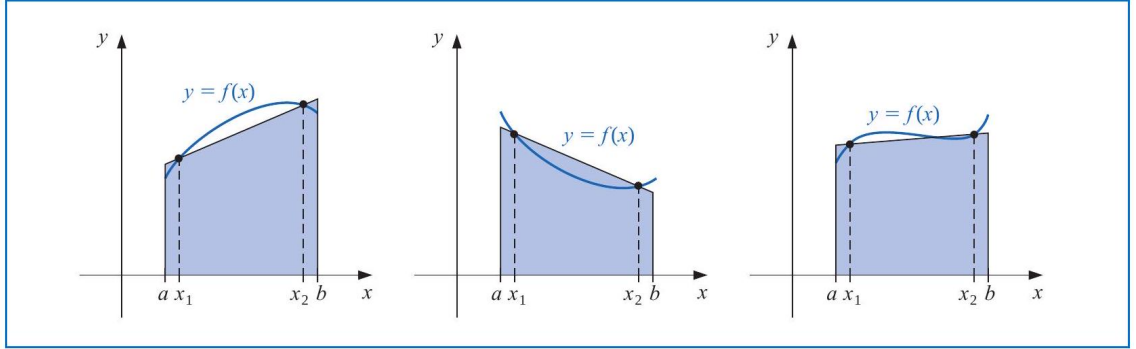


Figure 2: Figure 5

The coefficients $c_1, c_2, \dots, c_n$ in the approximation formula are arbitrary, and the nodes $x_1, x_2, \dots, x_n$ are restricted only by the fact that they must lie in $[a, b]$, the interval of integration. This gives us $2n$ parameters to choose. If the coefficients of a polynomial are considered parameters, the class of polynomials of degree at most $2n - 1$ also contains $2n$ parameters. This, then, is the largest class of polynomials for which it is reasonable to expect a formula to be exact. With the proper choice of the values and constants, exactness on this set can be obtained.

To illustrate the procedure for choosing the appropriate parameters, we will show how to select the coefficients and nodes when $n = 2$ and the interval of integration is $[-1, 1]$. We will then discuss the more general situation for an arbitrary choice of nodes and coefficients and show how the technique is modified when integrating over an arbitrary interval.

Suppose we want to determine c_1, c_2, x_1 , and x_2 so that the integration formula

$$\int_{-1}^1 f(x) dx \approx c_1 f(x_1) + c_2 f(x_2)$$

gives the exact result whenever $f(x)$ is a polynomial of degree $2(2) - 1 = 3$ or less, that is, when

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

for some collection of constants, a_0, a_1, a_2 , and a_3 . Because

$$\int (a_0 + a_1x + a_2x^2 + a_3x^3) dx = a_0 \int 1dx + a_1 \int xdx + a_2 \int x^2dx + a_3 \int x^3dx$$

this is equivalent to showing that the formula gives exact results when $f(x)$ is $1, x, x^2$, and x^3 . Hence, we need c_1, c_2, x_1 , and x_2 , so that

$$\begin{aligned} c_1 \cdot 1 + c_2 \cdot 1 &= \int_{-1}^1 1 dx = 2, & c_1 \cdot x_1 + c_2 \cdot x_2 &= \int_{-1}^1 x dx = 0 \\ c_1 \cdot x_1^2 + c_2 \cdot x_2^2 &= \int_{-1}^1 x^2 dx = \frac{2}{3}, & \text{and} \\ c_1 \cdot x_1^3 + c_2 \cdot x_2^3 &= \int_{-1}^1 x^3 dx = 0 \end{aligned}$$

A little algebra shows that this system of equations has the unique solution

$$c_1 = 1, \quad c_2 = 1, \quad x_1 = -\frac{\sqrt{3}}{3}, \quad \text{and} \quad x_2 = \frac{\sqrt{3}}{3},$$

which gives the approximation formula

$$\int_{-1}^1 f(x) dx \approx f\left(-\frac{\sqrt{3}}{3}\right) + f\left(\frac{\sqrt{3}}{3}\right) \quad (4.40)$$

This formula has degree of precision 3, that is, it produces the exact result for every polynomial of degree 3 or less.

Legendre Polynomials

The technique we have described could be used to determine the nodes and coefficients for formulas that give exact results for higher-degree polynomials, but an alternative method obtains them more easily. The set that is relevant to our problem is the Legendre polynomials, a collection $\{P_0(x), P_1(x), \dots, P_n(x), \dots\}$ with properties:

- (1) For each n , $P_n(x)$ is a monic polynomial of degree n .

Recall that monic polynomials have leading coefficient 1.

- (2) $\int_{-1}^1 P(x)P_n(x) dx = 0$ whenever $P(x)$ is a polynomial of degree less than n .

The first few Legendre polynomials are

$$\begin{aligned} P_0(x) &= 1, & P_1(x) &= x, & P_2(x) &= x^2 - \frac{1}{3}, \\ P_3(x) &= x^3 - \frac{3}{5}x, & \text{and} & & P_4(x) &= x^4 - \frac{6}{7}x^2 + \frac{3}{35}. \end{aligned}$$

The roots of these polynomials are distinct, lie in the interval $(-1, 1)$, have a symmetry with respect to the origin, and, most importantly, are the correct choice for determining the parameters that give us the nodes and coefficients for our quadrature method.

The nodes $x_1, x_2, \dots, x_n$ needed to produce an integral approximation formula that gives exact results for any polynomial of degree less than $2n$ are the roots of the n th-degree Legendre polynomial. This is established by the following result.

Theorem. Suppose that $x_1, x_2, \dots, x_n$ are the roots of the n th Legendre polynomial $P_n(x)$ and that for each $i = 1, 2, \dots, n$, the numbers c_i are defined by

$$c_i = \int_{-1}^1 \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} dx$$

If $P(x)$ is any polynomial of degree less than $2n$, then

$$\int_{-1}^1 P(x) dx = \sum_{i=1}^n c_i P(x_i)$$

The constants c_i needed for the quadrature rule can be generated from the equation in Theorem 1, but both these constants and the roots of the Legendre polynomials are extensively tabulated. Table 1 lists these values for $n = 2, 3, 4$, and 5 .

Example 1. Approximate $\int_{-1}^1 e^x \cos x dx$ using Gaussian quadrature with $n = 3$.

Solution: The entries in Table 1 give us

$$\begin{aligned} \int_{-1}^1 e^x \cos x dx &\approx 0.5e^{0.774596692} \cos 0.774596692 \\ &\quad + 0.8 \cos 0 + 0.5e^{-0.774596692} \cos(-0.774596692) \\ &= 1.9333904 \end{aligned}$$

Integration by parts can be used to show that the true value of the integral is 1.9334214 , so the absolute error is less than 3.2×10^{-5} .

Gaussian Quadrature on Arbitrary Intervals

An integral $\int_a^b f(x) dx$ over an arbitrary $[a, b]$ can be transformed into an integral over $[-1, 1]$ by using the change of variables (see Figure 3):

n	Roots $r_{n,i}$	Coefficients $c_{n,i}$
2	0.5773502692	1.0000000000
2	-0.5773502692	1.0000000000
3	0.7745966692	0.5555555556
3	0.0000000000	0.8888888889
3	-0.7745966692	0.5555555556
4	0.8611363116	0.3478548451
4	0.3399810436	0.6521451549
4	-0.3399810436	0.6521451549
4	-0.8611363116	0.3478548451
5	0.9061798459	0.2369268850
5	0.5384693101	0.4786286705
5	0.0000000000	0.5688888889
5	-0.5384693101	0.4786286705
5	-0.9061798459	0.2369268850

Table 1: Table 1

$$t = \frac{2x - a - b}{b - a} \iff x = \frac{1}{2}[(b - a)t + a + b]$$

This permits Gaussian quadrature to be applied to any interval $[a, b]$, because

$$\int_a^b f(x)dx = \int_{-1}^1 f\left(\frac{(b - a)t + (b + a)}{2}\right) \frac{(b - a)}{2} dt \quad (1)$$

Example 2. Consider the integral $\int_1^3 x^6 - x^2 \sin(2x)dx = 317.3442466$.

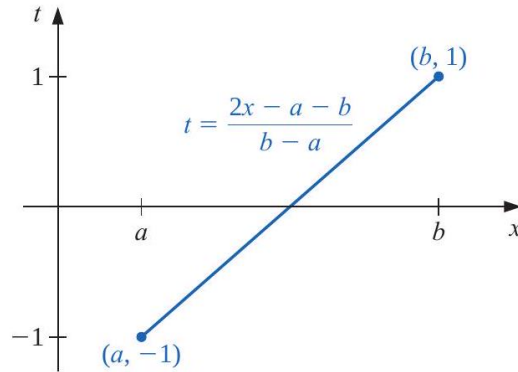


Figure 3: Figure 6

- (a) Compare the results for the closed Newton-Cotes formula with $n = 1$, the open Newton-Cotes formula with $n = 1$, and Gaussian Quadrature when $n = 2$.
- (b) Compare the results for the closed Newton-Cotes formula with $n = 2$, the open Newton-Cotes formula with $n = 2$, and Gaussian Quadrature when $n = 3$.

Solution:

- (a) Each of the formulas in this part requires 2 evaluations of the function $f(x) = x^6 - x^2 \sin(2x)$. The Newton-Cotes approximations are

$$\text{Closed } n = 1 : \frac{2}{2}[f(1) + f(3)] = 731.6054420$$

$$\text{Open } n = 1 : \frac{3(2/3)}{2}[f(5/3) + f(7/3)] = 188.7856682$$

Gaussian quadrature applied to this problem requires that the integral first be transformed into a problem whose interval of integration is $[-1, 1]$. Using Eq.1 gives

$$\int_1^3 x^6 - x^2 \sin(2x) dx = \int_{-1}^1 (t+2)^6 - (t+2)^2 \sin(2(t+2)) dt$$

Gaussian quadrature with $n = 2$ then gives

$$\int_1^3 x^6 - x^2 \sin(2x) dx \approx f(-0.5773502692 + 2) + f(0.5773502692 + 2) = 306.8199344;$$

- (b) Each of the formulas in this part requires 3 function evaluations. The Newton-Cotes approximations are

$$\text{Closed } n = 2 : \frac{(1)}{3}[f(1) + 4f(2) + f(3)] = 333.2380940$$

$$\text{Open } n = 2 : \frac{4(1/2)}{3}[2f(1.5) - f(2) + 2f(2.5)] = 303.5912023$$

Gaussian quadrature with $n = 3$, once the transformation has been done, gives

$$\int_1^3 x^6 - x^2 \sin(2x) dx \approx 0.5 f(-0.7745966692 + 2) + 0.8 f(2) \\ + 0.5 f(0.7745966692 + 2) = 317.2641516$$

The Gaussian quadrature results are clearly superior in each instance.